A Cost-Effectiveness Analysis of Prenatal Screening Strategies for Down Syndrome

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Objective: To evaluate which Down syndrome screening strategy is the most cost-effective.

Methods: Using decision-analysis modeling, we compared the cost-effectiveness of 9 screening strategies for Down syndrome: 1) no screening, 2) first-trimester nuchal translucency (NT) only, 3) first-trimester combined NT and serum screen, 4) first-trimester serum only, 5) quadruple screen, 6) integrated screening, 7) sequential screening, 8) integrated serum only, or 9) maternal age. Costs included cost of tests and resources used for raising a child with Down syndrome. One-way and multiway sensitivity analyses were performed for all model variables. The main outcome measures were cost per Down syndrome case detected, rate of delivering a liveborn neonate with Down syndrome, and rate of diagnostic procedure–related pregnancy loss for each strategy.

Results: Sequential screening detected more Down syndrome cases compared with the other strategies, but it had a higher procedure-related loss rate. Integrated serum screening was the most cost-effective strategy. Sensitivity analyses revealed the model to be robust over a wide range of values for the variables. The addition of the cost of genetic sonogram to the second-trimester strategies resulted in first-trimester combined screening becoming the most cost-effective strategy.

Conclusion: Within our baseline assumptions, integrated serum screening was the most cost-effective screening strategy for Down syndrome. If the cost of nuchal translucency is less than $57 or when genetic sonogram is included in the second-trimester strategies, first-trimester combined screening became the most cost-effective strategy.
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Level of Evidence: III

Many choices for aneuploidy screening are now available to women.[1] These screening tests can broadly be classified as first-trimester tests, second-trimester tests, or a combination of both. The first-trimester choices include sonographically measuring nuchal translucency (NT) alone, first-trimester serum screening (FSS) for which free β-human chorionic gonadotropin (hCG) and pregnancy associated plasma protein–A (PAPP-A) levels are assessed, or a combination of both serum and sonographic screening (NTSS).[2,3] The second-trimester choices include the quadruple serum screen (QUAD), which incorporates maternal serum α-fetoprotein (AFP), hCG, unconjugated estriol, and dimeric inhibin–A, the integrated combined test (ICS) which combines NTSS and the QUAD screen to calculate a single second-trimester risk estimate, and the integrated serum-only screen (ISS), which is similar to ICS, but only combines FSS and QUAD without NT. Finally, sequential screening (SEQ) can be performed in which NTSS and QUAD are performed *and reported* serially.[4] Although all these strategies incorporate maternal age, many clinicians consider maternal age alone a screening option.

With so many choices, patients and physicians are still uncertain regarding which screening option is optimal. Three previous economic analyses on Down syndrome screening have yielded conflicting results.[5–7] One of these concluded that the American practice at the time of using serum biochemistry was more cost-effective[5]; another reported NTSS as the most cost-effective strategy,[6] while one reported SEQ

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to be the most cost-effective.[7] None of these analyses considered the options of FSS or ISS, and the two earliest analyses did not include the ICS strategy. Our aim is to use a decision analytic approach to evaluate the most cost-effective strategy for Down syndrome screening in the prenatal period by assessing all of the most clinically prevalent testing options currently available to patients and physicians.

METHODS AND MATERIALS

We constructed a decision-analysis model to evaluate the cost-effectiveness of 9 separate testing options. The strategies included in the model were: 1) no screening, 2) NT only, 3) NTSS, 4) FSS, 5) QUAD screen, 6) ICS, 7) ISS, 8) SEQ, and 9) maternal age. In the maternal age strategy, women are offered a diagnostic test *only* if they are 35 years or older (standard of care in the United States). Table 1 is a glossary listing the abbreviations and description of the different strategies evaluated.

The baseline model consisted of pregnant women in the first and second trimesters of pregnancy. The analysis was performed from a societal perspective using relevant cost and outcomes related to each screening strategy in a theoretic cohort of 1 million pregnant women presenting before 11 weeks of gestation. The outcomes evaluated included the number of Down syndrome cases detected, procedure-related pregnancy loss rate, and cost-effectiveness ratio (cost per Down syndrome case averted) for each strategy.

We used the test characteristics, namely sensitivity and specificity, intrinsic to each screening method to estimate the chances of true- or false-positive and true- or false-negative results. We assumed that patients with negative genetic screen results, regardless of whether they are true or false, did not have invasive prenatal testing and would therefore have rates of a normal neonate or a Down syndrome neonate as potential outcomes. We assumed that patients with positive screening results, regardless of whether they are true or false, accepted invasive genetic testing with a constant acceptance rate (70%, range 30–100% for sensitivity analysis).

Probability estimates and plausible ranges for uncertain events for test characteristics (sensitivity and specificity), costs, and outcomes for each option were derived from a systematic and quantitative review of the English literature. Probability estimates were calculated by averaging point estimates in the literature, and the ranges for use in the sensitivity analyses were established using the extremes of the reported point estimates when available. See Tables 2 and 3 for the level of evidence for individual studies. We used a baseline first trimester prevalence of 1 in 595 for trisomy 21.[7] This baseline prevalence was adjusted using a spontaneous loss rate of 25% during the first trimester and 23% in the second trimester.[7] The sensitivities and specificities of the screening strategies are shown in Table 2.

For the model, we assumed a 70% (range 50–90%) uptake rate, or patient acceptance rate, for invasive procedures.[5–7,20,21] We also assumed a 70% pregnancy termination rate for Down syndrome (range 30–90% for sensitivity analysis). In the "maternal age" strategy, we assumed that women age 35 years or greater are given the option of having an invasive prenatal diagnostic test, that the sensitivity of maternal age for detecting Down syndrome is 30% (range 20–45% for sensitivity analysis). The model assumed that 13% of the pregnant population will be aged 35 years or more.[24] The age distributions were varied in the sensitivity analysis to account for the increasing proportion of women of advanced maternal age getting pregnant.

The cost estimates and utilities used in the model for the baseline analysis are shown in Table 3. The cost estimates were derived from the literature and adjusted for 2004 U.S. dollars using yearly inflation rates. When available, the Medicare National Fee was used for the cost estimates and when unavailable, the hospital charges were multiplied by a cost-to-charge ratio (0.6) for the Commonwealth of Pennsylvania (to

Table 1. Glossary of Abbreviations for Down Syndrome Screening Strategies Described in the Manuscript

Abbreviation for Screening Strategy (meaning)	Description of Strategy
FSS (First-trimester serum screening)	First-trimester serum only
NT (Nuchal translucency)	Nuchal translucency only
NTSS (Nuchal translucency and serum screening)	Nuchal translucency + first-trimester serum screening
ICS (Integrated screening)	NTSS + QUAD* (combined result given once in second trimester)
ISS (Integrated serum screening)	FSS + QUAD*
SEQ (sequential screening)	NTSS + QUAD* (results given serially in first and second trimesters, respectively)
QUAD (Quadruple test)	α-fetoprotein; unconjugated estriol; β-hCG and dimeric inhibin–A

* β-hCG is performed only once in either trimester.

Table 2. Baseline Probabilities, Sensitivity, and Specificity of Screening Strategies Included in the Decision Tree

Strategy	Baseline Estimate (%)	Range Used in Sensitivity Analysis (%)	Reference	Level of Evidence*
NT only	77	29–82	8–13	II-2 and III
Specificity NT only	94	83–98	8–13	II-2 and III
Sensitivity NTSS	82	80–91	3,10,14–16	II-2
Specificity NTSS	95	91–96	3,10,14–16	II-2
Sensitivity FSS	60	30–75	6,17	II-2
Specificity FSS	95	90–96	6,17	II-2
Sensitivity QUAD	70	67–76	16,18	II-2
Specificity QUAD	95	90–96	16,18	II-2
Sensitivity integrated (ICS)	94	85–96	4,7,16	II-2
Specificity integrated (ICS)	95	91–98	4,7,16	II-2
Sensitivity integrated serum (ISS)	85	75–90	4,16	II-2
Specificity integrated serum (ISS)	97	95–99	4,16	II-2
Sensitivity sequential (SEQ)	95	80–96	4,7,16	II-2
Specificity sequential (SEQ)	90	85–95	4,7,16	II-2
Sensitivity of genetic sonogram	69	30–90	7,19	II-2
Specificity of genetic sonogram	92	80–100	7,19	II-2

NT, nuchal translucency; FSS,first-trimester serum screening; NTSS, complete first-trimester screening; QUAD, quadruple screen; ISS, integrated serum screening; ICS, integrated combined screening; SEQ, sequential screening.
* U.S. Preventive Task Force. Am J Prev Med 2001;20(Suppl 1):21–35. Level I is the best evidence level and III is the worst evidence level.

Table 3. Costs Included in the Decision Tree of Screening Options for Down Syndrome

Variable	Baseline Estimate (US$)	Range (US$)	Reference	Level of Evidence*
Cost of NT only	57	35–150	6; see text	II-2
Cost of FSS	75	50–170	6; see text	II-2
Cost of Quad	147	50–200	Local sources; see text	NA
Cost of ICS	204	100–370	7	II-2
Cost of sequential (SEQ)	279	150–350	Local sources and 7	NA and II-2
Cost Integrated serum (ISS)	222	100–370	Local sources; see text	NA
Cost of CVS (counseling, US, procedure and laboratory)	890	600–1200	5,6,25	II-2 and III
Cost of amniocentesis (counseling, US, procedure and laboratory)	880	600–1200	5,6,25,26	II-2 and III
Cost first-trimester pregnancy termination	600	400–800	5,6,25,26	II-2 and III
Cost second-trimester pregnancy termination	1200	900–1500	5,6,25,26	II-2 and III
Cost of genetic sonogram	148	80–200	Local sources; see text	NA
Cost of Down syndrome	612,150	400,000–800,000	5,6,7; see text	II-2

NT, nuchal translucency; FSS,first-trimester serum screening; NTSS, complete first-trimester screening; QUAD, quadruple screen; ISS, integrated serum screening; ICS, integrated combined screening; SEQ, sequential screening.
* U.S. Preventive Task Force. Am J Prev Med 2001;20(Suppl 1):21–35. Level I is the best evidence level and III is the worst evidence level.

account for the typical scenario in which charges tend to be higher than Medicare reimbursements).[7,25,26] For example, as there are no published costs for NT, we reduced our local charge by a factor of 0.6 to obtain the estimated "cost" of $57.

The final cost of a strategy will be a sum of the different components of that pathway. For example, the cost of the NTSS strategy resulting in a normal karyotype after a chorionic villus sampling (CVS) will be derived by the formula (cost of NT + cost of FTSS + cost of CVS). The costs were all discounted at a rate of 3%. Discounting is performed in economic analysis to account for the fact that individuals have a preference for earlier consumption compared with later consumption. To recognize this difference in preferences, utilities and costs are usually discounted by 3–5%.

Women who are screen positive for Down syndrome have two possible options, including doing nothing or having an invasive diagnostic procedure. The diagnostic procedures include a CVS in the first trimester or an amniocentesis in the second trimester. If the diagnostic test result is positive, the woman is faced with the options of pregnancy termination or continuing the pregnancy. The potential harmful effects from any of these options include a diagnostic procedure–related pregnancy loss and complications from pregnancy termination. We have included these possible consequences in our outcome analyses. We estimated a procedure-related fetal loss rate of 0.75% after amniocentesis and 1.4% after CVS.[6,7,22,23] These estimates were varied widely for our sensitivity analysis (0.5–2% for CVS and 0.03–1.0% for amniocentesis).

In the baseline analysis, pathway probabilities were used to calculate the cost, number of Down syndrome cases detected, and the procedure-related loss rate for each strategy. A separate model was designed incorporating the cost of genetic sonogram into the second trimester (ICS, ISS, SEQ, and QUAD) strategies. For this analysis it was assumed that 60% of women with abnormal screening will receive a genetic sonogram.

We performed one-way and multi-way sensitivity analyses on all model variables, commensurate with the degree of uncertainty for each point estimate. The ranges used for the sensitivity analysis are shown in Tables 2 and 3. We used a commercially available decision-analysis software package (TreeAge Pro 2004, Tree Age Inc, Williamstown, MA) to perform all computations.

RESULTS

Under the baseline assumptions, the model favors the ISS as the most cost-effective Down syndrome screening strategy. The ISS was the least expensive strategy per Down syndrome cases averted. Although the options of NT, FSS, NTSS, and QUAD cost less, these four strategies had lower Down syndrome detection rates and higher numbers of procedure-related losses compared with ISS. When we repeated the analysis using the QUAD screen strategy (standard of care for most U.S. centers, but not the cheapest strategy in the model) as the baseline for comparison, ISS remained the most cost-effective strategy. The results of the analysis evaluating clinical outcomes and costs are shown in Table 4. All strategies considered were cheaper compared with no screening. Changing from the QUAD screen strategy to NTSS will result in 98 additional cases of Down syndrome averted and a societal savings of $8,516 per additional case averted. If the change is from QUAD to ISS, 123 additional Down syndrome will be averted at a cost of $104 per case averted. The "maternal age" strategy was the most costly per Down syndrome case averted.

When the cost of genetic sonogram was included in the second trimester strategies, NTSS became the most cost-effective strategy. The results of this analysis showed that the total population cost of ISS increased

Table 4. Clinical Outcomes and Costs From Baseline Analysis of the Cost-Effectiveness of Down Syndrome Screening (Theoretical Cohort of 1 Million Pregnancies)

Screening Option	No. of Down Syndrome Cases Detected (Unadjusted No.)	No. of Down Syndrome Live Birth Averted*	Procedure-Related Euploid Losses	Down Syndrome Case Averted/ Euploid Losses	Total Population Cost of Strategy in 2004 (Million US$)	Cost per Down Syndrome Case Averted (US$)
NT only	906† (1,133)	634	588	1.1	175	27,603
FSS	706† (883)	494	490	1.0	183	37,045
NTSS	965† (1,206)	675	490	1.9	242	35,851
QUAD	824	577	263	2.2	256	44,367
ISS	1,000	700	158	4.4	312	44,571
ICS	1,106	774	263	2.9	389	50,258
SEQ	1,118	783	980	0.80‡	429	54,789
Maternal age ≥ 35 y	353	247	490	0.50‡	78	315,789
No screening	–	–	–	–	1,029	–

NT, nuchal translucency; FSS, first-trimester serum screening; NTSS, complete first-trimester screening; QUAD, quadruple screen; ISS, integrated serum screening; ICS, integrated combined screening; SEQ, sequential screening.
* Assuming a 70% termination rate.
† Adjusted for a 25% rate of spontaneous loss between first trimester and second trimester.
‡ Associated with more losses than Down syndrome averted.

to $316 million, and the cost per case of Down syndrome averted increased to $45,719. Similar increases were noted for the QUAD, ICS, and SEQ strategies with the NTSS cost remaining similar to the baseline analysis.

Although ISS was the most cost-effective strategy, the SEQ strategy had a higher number of Down syndrome cases detected compared with the other strategies. Sequential screening was associated with the detection of 1118 cases of Down syndrome which is 26% higher that the QUAD. However, SEQ results in a substantially higher number of procedure-related euploid pregnancy losses (Table 4), rendering this strategy less desirable and effective overall. The ratio of Down syndrome cases averted to euploid pregnancy losses was less than 1.0 for the SEQ strategy and "maternal age" compared with ISS which had the best ratio of 4.4.

Sensitivity analyses were performed by varying each estimate of the sensitivity, specificity, outcome probabilities, utilities, and costs within their plausible ranges as shown in Tables 2 and 3. The sensitivity analyses revealed the model results to be stable over a wide range of values for included variables. There was a reduction in the cost-effectiveness ratio for ISS when the cost of second-trimester serum was below $125, but ISS remained the most cost-effective strategy. The model was sensitive to a cost of NT below $57. Below this cost NTSS was more cost-effective.

DISCUSSION

Under our baseline assumptions, the ISS strategy was the most cost-effective screening strategy for Down syndrome. Our analysis also revealed that although sequential screening was associated with a higher detection rate for Down syndrome, this strategy resulted in a dramatically higher number of procedure-related euploid pregnancy losses. This higher number of procedure-related pregnancy losses is not surprising because serial screening allows multiple points in time (in this case, two) that a positive result can occur, prompting a diagnostic procedure. Unfortunately, most of the positive test results are falsely positive causing the patient to incur a higher pregnancy loss rate in the absence of true fetal disease. The analysis also showed that changing from QUAD screen to NTSS, ISS, or ICS will result in higher numbers of Down syndrome cases averted and, in the case of NTSS, may result in societal cost savings. The ratio of Down syndrome cases averted to the euploid pregnancy losses was clearly in favor of ISS.

Although the genetic sonogram has been shown in several studies to be beneficial in avoiding unnecessary diagnostic procedures,[19] the primary aim of our analysis was to compare the extensive array of screening options available before reporting for a genetic sonogram. We however, performed a second analysis to reflect the importance of the genetic sonogram. The analysis revealed that the addition of the cost of genetic sonogram to the women with positive second-trimester screening tests resulted in NTSS becoming the most cost-effective strategy.

Our baseline findings differ from previously published cost-effectiveness studies on Down syndrome screening.[5–7,27] Vintzileos et al[5] compared the screening strategies in Britain (NT only) with the practice in the United States at that time (second-trimester serum) and concluded that the U.S. strategy was more cost-beneficial. Since that report, practice patterns in the United States has changed with the introduction of first-trimester screening options, making their conclusions less generalizable. Caughey et al[6] reported that first-trimester screening using NT and serum (NTSS) was the most cost-effective strategy. Conversely, Biggio et al[27] found SEQ to be the most cost-effective strategy, but they also noted that this strategy was associated with the highest miscarriage rate.[7] Gilbert et al[27] concluded that the four most cost-effective strategies were ICS, NT alone, NTSS, and QUAD screen. The last study was a cohort evaluating the National Health Service in the United Kingdom and did not evaluate SEQ or ISS. In addition, the estimates used in the analysis by Gilbert et al[27] were from one study using stored serum samples to evaluate the characteristics of screening tests only, and the cost of NT was highly underestimated (4 sterling pounds).[27]

We used Medicare costs in our analysis rather than charges as some investigators had used in previous studies.[5,6] Using the Medicare National Fee is a more accurate estimate of cost of the service compared with using hospital charges and is, therefore, more representative of the societal burden.[7,25,28] In this analysis we have evaluated the outcomes from a societal perspective as well as the individual patient's perspective. In addition our "maternal age" strategy incorporated the use of advanced maternal age ($\geq$ 35 years) as a screening technique to more closely reflect current clinical practice. Biggio et al[7] however limited their analysis to women aged less than 35 years old.

The results of our baseline analysis were not sensitive to one-way or two-way sensitivity analyses for all variables, except for the cost of NT. However, we used a conservative estimate of the cost of NT in our model because it is currently not listed in the Medicare National Fee. Because it is highly unlikely that the cost of performing NT (a time-consuming and sometimes technically challenging procedure) will be below $57, the conclusions from our analysis can be considered robust. One important advantage of the

ISS option is that it avoids the technical and logistical difficulties of NT measurements, and it is easier to perform biochemical quality assurance compared with the NT certification process. The major disadvantage of the ISS (similar to ICS) strategy is that results are held until the second trimester (an average of 3–4 weeks), which may be unacceptable to some women.

We must emphasize that our analysis, like most cost analyses involving prenatal screening and diagnosis, is based on some underlying assumptions that the clinician or patient must consider in deciding which screening strategy to use.[29] For example if the society sets a willingness-to-pay threshold at $50,000 per Down syndrome case averted, then most of the strategies could be considered cost-effective. In addition, there are other benefits of NT-based screening, such as the detection of cardiac defects that cannot be captured by this cost-effectiveness analysis.

Our analysis has some limitations. There is limited information on the sensitivity and specificity of options such as FSS, ISS, ICS, and SEQ because they are relatively new screening tests evaluated with few population-based studies. The sensitivity of biochemical screening tests is higher with reliable pregnancy dating using ultrasonography.[30] Our model did not include cost for ultrasound dating in the integrated serum strategy because this is not a universal practice in the United States. This could have biased our results in favor of these biochemical screening strategies. Our analysis was limited to Down syndrome screening, and the results cannot be generalized to all trisomies. Although our analysis was from a societal perspective, certain indirect costs such as loss of work time, physician time, counseling time, and other intangible costs could not be captured by our model. In summary, this cost-effectiveness analysis determined that the integrated serum screening strategy is the most cost-effective. However, when considered from the perspective of contemporary clinical practice, which incorporates the cost of genetic ultrasonography, NTSS became the most cost-effective.

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